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Plasma proteomics identify novel biomarkers and dynamic patterns of biological aging



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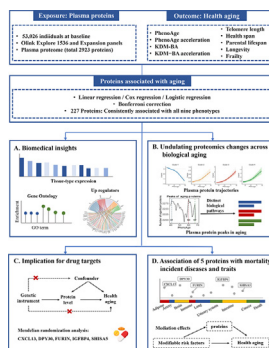
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HIGHLIGHTS

- 227 plasma proteins were linked to aging across 9 traits, involving inflammation and regeneration pathways.
- Plasma proteome shows age-related shifts, peaking at ages 41, 60, and 67.
- Inflammatory and regenerative pathways are key components in aging.
- MR supports causal links between CXCL13, DPY30, FURIN, IGFBP4, SHISA5, and aging.
- Aging-related proteins affect mortality and cardiovascular outcomes.

GRAPHICAL ABSTRACT

Upper Panel: We leveraged plasma proteomic data (2,923 proteins) from 53,026 participants at baseline. Health aging-related traits were measured at baseline and online follow-up, including: PhenoAge, PhenoAge acceleration, KDM-BA, KDM-BA acceleration, telomere length, healthspan, parental life span, longevity, and frailty. We first conducted an unbiased proteome-wide association analysis to identify related proteins using Cox, linear, and logistic regression models. **A.** The biological functions of the identified proteins were explored through various analyses, such as tissue/cell expression analysis, ontology pathway, and up-regulator enrichment analysis, to gain insights into their potential roles and molecular mechanisms in aging. **B.** We characterized the trajectories of plasma proteins during aging, discerning both linear and non-linear patterns. Additionally, we identified essential time periods during aging, and the proteins at these time periods had distinct biological pathways. **C.** Then, we employed Mendelian randomization analysis to investigate potential causal relationships between plasma proteins and health aging, aiming to uncover potential drug targets for aging. **D.** Next, we investigated the associations of the proteins with traits, mortality, or diseases. The key proteins as mediating factors in the pathway between modifiable risk factors and aging.



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ABSTRACT

Introduction: Plasma proteomics examines levels of thousands of proteins and has the potential to identify clinical biomarkers for healthy aging.

Objectives: This large proteomics study aims to identify clinical biomarkers for healthy aging and further explore potential mechanisms involved in aging.

Methods: This study analyzed data from 51,904 UK Biobank participants to explore the association between 2,923 plasma proteins and nine aging-related phenotypes, including PhenoAge, KDM-Biological Age, healthspan, parental lifespan, frailty, and longevity. Protein levels were measured using proteomics, and associations were assessed with a significance threshold of $P < 1.90E-06$. We utilized the DE-SWAN method to detect and measure the nonlinear alterations in plasma proteome during the process of biological aging. Mendelian randomization was applied to assess causal relationships, and a PheWAS explored the broader health impacts of these proteins.

Results: We identified 227 proteins significantly associated with aging ($P < 1.90E-06$), with the pathway of inflammation and regeneration being notably implicated. Our findings revealed fluctuating patterns in the plasma proteome during biological aging in middle-aged adults, pinpointing specific peaks of biological age-related changes at 41, 60, and 67 years, alongside distinct age-related protein change patterns across various organs. Furthermore, mendelian randomization further supported the causal association between plasma levels of CXCL13, DPY30, FURIN, IGFBP4, SHISA5, and aging, underscoring the significance of these drug targets. These five proteins have broad-ranging effects. The PheWAS analysis of proteins associated with aging highlighted their crucial roles in vital biological processes, particularly in overall mortality, health maintenance, and cardiovascular health. Moreover, proteins can serve as mediators in healthy lifestyle and aging processes.

Conclusion: These significant discoveries underscore the importance of monitoring and intervening in the aging process at critical periods, alongside identifying potential biomarkers and therapeutic targets for age-related disorders within the plasma proteomic landscape, thus offering valuable insights into healthy aging.

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Introduction

The gap between health and lifespan suggests the potential for actively seeking treatment methods. The Geroscience Hypothesis, posits that aging serves as the primary risk factor for all chronic diseases [1]. Therefore, targeting factors associated with aging may offer greater advantages in improving health span compared to addressing each chronic disease individually. However, the current therapeutic interventions for aging are not ideal. The limited success in achieving sustained remission is closely linked to our incomplete understanding of its pathogenesis. Unraveling these elusive mechanisms is essential to lay the groundwork for more effective therapeutic interventions. In this context, proteins serve as the ultimate products of gene expression, playing a pivotal role in cellular function and offering direct insights into the pathophysiology of diseases [2]. Therefore, analyzing protein dysregulation in age-related phenotypes may reveal the internal processes preceding aging. This provides insights into potential mechanisms and paves the way for personalized treatment strategies [3].

The previous emphasis on individual proteins in aging research has indeed yielded the discovery of several proteins related to aging. For example, cyclin-dependent kinase inhibitor P16INK4A is involved in cellular senescence [4], proinflammatory cytokine IL-6 is associated with inflammation [5], and insulin-like growth factor IGF-1 is linked to longevity pathways [6]. Additionally, APOE [7] and FOXO3 [8] have been identified as factors influencing lifespan and longevity. Through comprehensive proteomic analysis, a more thorough understanding of the aging process can be achieved, potentially leading to the identification of additional biomarkers associated with it. However, several unresolved issues persist regarding proteomic studies on aging. Some previous studies focused on protein traits linked to chronological age in plasma but were limited by small samples and a bias towards age-related proteins over aging phenotypes [9,10]. Although chronological

aging progresses uniformly, biological aging varies among individuals due to physiological and disease-related changes, resulting in diverse health outcomes [11]. Therefore, biological age, which reflects an individual's overall physiological state, offers a more comprehensive assessment [12]. Investigating the dynamics of plasma proteome during aging process is essential for revealing the molecular mechanisms associated with age-related disorders. Furthermore, conducting proteomic studies across various phenotypes enhances our understanding of the biological mechanisms and molecular markers linked to human aging [13]. Proteomics enables the identification of aging-associated proteins through single-phenotype approaches, such as extreme longevity, healthspan, and familial lifespan. Currently, genome-wide association studies (GWAS) focus on single phenotypes to explore intervention targets in proteomics [14,15]. However, these methods overlook the shared biology among these traits and other aging-related factors, such as proteomic markers of aging and physiological frailty. Investigating multiple phenotypes allows for a more comprehensive exploration of the underlying biological mechanisms of human aging and aids in the identification of more precise biomarkers for promoting healthy aging [16]. Bridging this knowledge gap also provided possibilities for the early identification and subsequent intervention of age-related disorders.

Therefore, in the prospective UK Biobank (UKB) cohort, we investigated the relationship between 2923 baseline plasma protein levels and healthy aging in a sample of over 50,000 participants. Initially, we employed linear, logistic, and cox regression analyses to examine the proteomic concentrations associated with aging to identified plasma proteins associated with biological age. Additionally, we characterized their biological functions and cellular expressions in the aging and identified three important plasma proteomic peaks of biological aging during the profiling of plasma proteome trajectories during biological aging. To underscore the clinical implications, we conducted Mendelian randomization

analysis to establish causal links, spotlighting potential therapeutic targets and discovered clinical implications for health related to these proteins.

Methods

Study population

The UKB is a ongoing cohort study that commenced in April 2007 and has enrolled over 500,000 individuals aged 40 to 69 from various locations in the UK [17]. Participants visited assessment centers where they completed questionnaires, underwent physical examinations, and provided biological samples [18]. The study was ethically approved and participants provided written consent. This research made use of the UKB resource in accordance with the terms of approved application number 19542.

Proteomics measurement

From the over 500,000 UKB participants, plasma samples collected during the baseline assessment from more than 50,000 individuals were selected. Among these, 46,673 individuals were randomly chosen, 1,268 were selected due to their participation in the COVID-19 repeat imaging study, and 6,365 were selected by thirteen participating consortium members based on specific research interests. These selected samples are generally representative of the overall UKB population. For detailed participant selection methodology and sample handling, refer to the [Supplementary Information](#) by Sun et al [19]. Blood samples were collected in EDTA tubes, centrifuged at 2500g for 10 min at 4 °C to isolate plasma, and stored at –80 °C. Samples were then sent to Sweden for analysis using the Olink Explore™ Proximity Extension Assay, quantifying 2,923 protein analytes corresponding to 2,941 proteins across various panels. Stringent quality control ensured inter- and intra-panel coefficients of variation below 20 % and 10 %, respectively. Further details on sample selection, processing, and quality control are available in prior publications [20]. Protein levels were reported as Normalized Protein eXpression (NPX) values, normalized against expansion controls and log-transformed to minimize variations without additional processing by UKB.

Outcome definitions

The selected aspects related to human aging encompass a range of factors, including biological age, telomere length, healthspan, parental lifespan, exceptional longevity, and frailty. Further details can be found in Supplementary Table 2. We quantified biological age (BA) using composite measures from blood chemistry and clinical data, applying the Klemmer-Doubal method (KDM) and PhenoAge (PA) [21]. Previous studies have detailed the calculation and interpretation of these measures [22,23]. KDM involves regressions of biomarkers against chronological age (CA), which indicates the age at which an individual's physiology matches the average physiology observed in the US National Health and Nutrition Examination Surveys (NHANES) III dataset. PhenoAge is based on a mortality prediction score derived from biomarkers and CA, which reflects the age at which the average mortality risk in NHANES III is equal to the predicted risk. We used the BioAge software package for the calculations [23]. To measure the difference between BA and CA, regressions of CA from KDM-BA and PA were conducted, treating the resulting values as “age acceleration”.

The UKB employed polymerase chain reaction (PCR) methods to measure telomere length (LTL), with the results being blinded to phenotypic data. Comparison was made between the amplification of the telomere length PCR product (T) and that of a PCR product from a reference gene (S), resulting in a T/S ratio. The T and S values

were calibrated against a calibrator sample (DNA pooled from 20 individuals) in each run. This study employed a log-transformed, z-score-adjusted T/S ratio to accurately represent relative LTL, while accounting for technical factors [24].

Healthspan was defined as the time at the first occurrence of any of eight predefined healthspan-ending events, following the methodology of the published UKB healthspan study [25]. These events included seven age-related morbidities—congestive heart failure (CHF), myocardial infarction (MI), chronic obstructive pulmonary disease (COPD), stroke, dementia, diabetes, and cancer—as well as death. Healthspan was defined as the absence of these diseases in the study sample. Participants were considered to have reached the end of their healthspan at the earliest diagnosis of any of these conditions or at the time of death.

Parental life span was calculated based on participants' self-reports of their parents' ages at the time of death [26]. In cases where data on the life expectancy of one parent was not available, the life expectancy of the other parent was substituted. In the absence of information on the life expectancy of one parent, the average life expectancy of both parents was calculated. Individuals were excluded from the analysis if both parents were alive or if their age was less than 40 years, as these cases were likely to indicate accidental deaths.

The exceptional longevity of participants was determined by whether at least one parent lived to age 85 or older, or if neither parent reached age 85 [27]. Furthermore, participants were classified according to the age of their mother and father, whether deceased or alive. The term “maternal longevity” was used to indicate that the mother had reached the age of 85 or older, regardless of the father's status. Similarly, paternal longevity was defined as the father reaching 85 years of age or older, irrespective of the mother's status.

To assess frailty status, we utilized the Fried phenotype, which incorporates 5 self-reported or objectively measured components: weight loss, exhaustion, low grip strength, physical inactivity, and slow walking pace [28]. Given that the specific measurements and data collection protocols in the UK Biobank differed from those used in the Cardiovascular Health Study (CHS), we modified the original phenotype criteria to ensure compatibility with the available UK Biobank data [29]. Supplementary Table 2 presents our revised definitions alongside those established by Fried and colleagues. The participants were categorized as robust (not meeting any of the Fried criteria), pre-frailty (meeting 1–2 criteria), or frailty (meeting ≥ 3 criteria) based on the availability of complete data [28].

Identification of aging-related proteins

The Cox regression model was used to assess the relationship between protein levels and healthspan. Linear regression models were used to examine the association between protein levels and PhenoAge, PhenoAge acceleration, KDM-BA, KDM-BA acceleration, telomere length, and parental life span. Logistic regression models were used to examine the association between protein levels and frailty and longevity. Potential confounders were adjusted, including baseline chronological age, sex, ethnicity, Townsend's deprivation index (TDI), education level, smoking status, alcohol status. We applied a stringent Bonferroni correction ($P < 0.05/2,923/9$) to evaluate significant associations for each outcome. Significant associations were determined when two-tailed p-values after Bonferroni correction were less than 0.05.

Functional enrichment analysis

Proteins included in the subsequent analysis ($n = 227$) met the following criteria: they were associated with all nine phenotypes,

and exhibited consistent associations across phenotypes. Tissue enrichment analysis was performed using the GENE2FUNC function of FUMA. DEG sets were first identified by two-sided t-tests for each tissue against the 54 tissue types in the GTEx database. Hypergeometric tests were then used to explore aging-associated proteins within these DEG sets. To explore the functional enrichment of the significant proteins identified, we utilized the Enrichr platform (<https://maayanlab.cloud/Enrichr/>), using the complete set of proteins from the Olink panel as the background gene set [30]. The enrichment analysis was performed based on Gene Ontology (GO) terms and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways. To control for multiple testing, the Benjamini-Hochberg method was applied, and results with a false discovery rate (FDR) below 0.05 were considered statistically significant. We used the TRRUST database, which contains 8444 TF-target relationships involving 800 human transcription factors, to investigate the upstream regulators of aging-associated proteins [31]. This analysis, combined with Metaspace, helped identify potential transcription factors responsible for regulating the expression and activity of these proteins. The top representative biological processes, pathways and TF were then visualized using the R package ggplot2.

Protein trajectories during biological aging

We selected 227 proteins nominally associated with aging to estimate their trajectories during biological aging. Initially, we z-normalized the protein levels. Then, we employed locally estimated scatterplot smoothing (LOESS) regression [32] with a span of 300 to fit each plasma protein with biological age, considering all covariates used in the association analysis. Subsequently, we calculated pairwise differences between LOESS estimates using Euclidean distance. Finally, we conducted hierarchical clustering using the complete method to categorize different trajectory patterns into four clusters across the aging process.

Differential expression – sliding window analysis (DE-SWAN)

We utilized the DE-SWAN method, implemented through the R package DESwan, to detect and measure the nonlinear alterations in plasma proteome during the process of biological aging [33]. Initially, we roughly estimated the biological age of healthy participants and included only those aged between 40 and 70 based on the distribution of biological ages. This resulted in the utilization of 30 centers, each with a ± 1 -year window. For the analysis of differential expression, we employed the following model:

Protein level $\sim \alpha + \beta_1$ biological age low/high + β_2 sex + β_3 ethnicity + $\dots + \epsilon$.

The significance levels, represented by q-values for each wave of biological age, were determined using Benjamini-Hochberg correction. Additionally, Type II sum of squares was computed using the ANOVA function within the R package car. Proteins exhibiting q-values < 0.05 were considered statistically significant. The corresponding biological pathways of each cluster were further inferred with KEGG databases, as previously described [34]. Only biological processes with p-values < 0.05 were considered significant. Based on the tissue enrichment results from GTEx, we identified proteins associated with various organs. Our analyses were focused on specific tissues, including adipose tissue, arteries, the urinary system (including the kidney and bladder), the brain, the upper gastrointestinal tract (including the esophagus and stomach), immune tissue, the intestine, and the lungs. These were selected due to their well-understood roles in age-related diseases and the presence of relevant phenotype data in the cohorts we examined. The DE-SWAN method was employed to identify and quantify

the nonlinear alterations in the plasma proteome of each organ during the biological aging process.

Mendelian randomization and drug ability assessment

We employed two-sample MR to explore proteins' causal impact on healthy aging, using them as the exposure variable and assessing their effect on healthy aging as the outcome. GWAS summary data for 9 healthy aging-related phenotypes were internally computed using UKB data, excluding participants with protein data. We focused on European participants from the UKB for pQTL mapping. Genetic instruments for proteins were identified by significant SNPs ($p < 5E-08$) that underwent clumping (1000 kb distance, LD $r^2 \leq 0.01$) using the European LD reference panel from the 1000 Genomes Project. We applied Steiger filtering, excluding SNPs with palindromic features and intermediate allele frequencies, and used the inverse-variance weighted (IVW) method for causal estimates in our main analyses. The association between instrumental variables (IVs) and risk factors was assessed using R² and F statistics. Higher values indicate greater instrument strength and statistical power in the MR analysis. Statistical analyses were performed using the R package "TwoSampleMR" [35]. The druggability of each prioritized target was checked according to Finan's criteria [36].

Phenome-wide association analysis

We conducted a wide-ranging analysis to explore the connection between key healthy aging-related proteins ($n = 5$) and various phenotypes. Based on pathway enrichment and tissue expression analysis results, six categories of phenotypes were chosen for further study: body composition measures, blood lipid, renal function, brain functions, immune, and lung function measures. This study primarily focuses on arterial diseases, urinary system disorders, brain-related illnesses, upper gastrointestinal tract disorders, intestinal disorders, lung diseases, cancer, and deaths. Diagnoses were made using the ICD-10 coding system, with individuals having pre-existing disorders at baseline excluded from analysis. The follow-up endpoint was defined as the earliest occurrence of a disorder, date of death, or conclusion of the follow-up period (June 30, 2023). Cox proportional model assessed the associations between aging-associated proteins and incident disorders, while linear regression model identified associations between aging-associated proteins and phenotypes. The model adjusted for potential confounding factors such as chronological age, sex, ethnicity, TDI, education level, smoking status, and alcohol status, and Bonferroni correction was applied for multiple corrections. We constructed triangular graphs linking modifiable risk factors, proteins, and healthy aging traits, all of which exhibited consistent association directions as hypothesized. Only five proteins, which were identified as MR positive, were included in the mediation analysis to explore the potential mediating role of the proteins between risk factors and healthy aging. Additional details can be found in Supplementary Methods and Supplementary Table 17.

Results

Population characteristics

A total of 51,904 individuals from the UKB, aged between 39 and 70 years, were enrolled in the study, of whom 46 % were men (Supplementary Table 1). Plasma samples were processed using the Olink Explore 3072, resulting in 2,923 unique proteins (Supplementary Table 3). A subset of participants who provided data on healthspan, telomere length (LTL), parental lifespan, excep-

tional longevity, biological age, and frailty were included in the analysis (Supplementary Table 2). The overall study design is shown in Graphical Abstract.

Proteome-wide association of aging

A total of 26,307 associations were tested, covering 2,923 proteins and 9 outcomes. In the whole dataset, a total of 12,193 significant associations surpassed the multiple comparison tests of the Bonferroni correction ($P < 1.90E-06 \approx 0.05/2,923/9$). The study found that KDM-BA had the highest number of associated proteins ($n = 2,188$, 74.85 %), followed by PhenoAge ($n = 2,168$, 74.17 %), frailty ($n = 939$, 32.12 %), healthspan ($n = 830$, 28.40 %), LTL ($n = 820$, 28.05 %), longevity ($n = 482$, 16.49 %), and parental life span ($n = 410$, 14.03 %). We assessed the quantity and proportion of proteins significantly associated with 9 health-related aging phenotypes. The largest quantity of proteins was associated with inflammation, except for telomere length and parental life span (PhenoAge, $n = 573$; PhenoAge acceleration, $n = 573$; KDM – BA, $n = 596$; KDM-BA acceleration, $n = 596$; healthspan, $n = 253$; longevity, $n = 151$; frailty, $n = 279$). Cardiometabolic proteins followed in quantity, while proteins related to neurology and oncology exhibited similar quantities across multiple phenotypes (Fig. 1B). Further details of the results are available in Fig. 1 and Supplementary Table 4.

We found that 227 proteins were consistently associated with all phenotypes. Of these, 2 were protective against aging (PON3 and UMOD), while 225 were promoting aging. We found that PON3 and UMOD showed significant protective associations in all nine aging-related phenotypes (Supplementary Table 4). PON3, a paraoxonase family member, binds high-density lipoprotein (HDL) in circulation and inhibits low-density lipoprotein (LDL) oxidation through its antioxidant activity, thereby attenuating oxidative stress and atherosclerosis progression—key drivers of cardiovascular aging [37,38]. UMOD, the predominant urinary protein, is critical for renal tubular integrity, defense against urinary infections, and systemic inflammation modulation [39,40]. Genetic variants in UMOD are linked to renal aging trajectories and healthspan [41]. The protective effects of both proteins across diverse aging-related phenotypes suggest that they may systemically delay aging processes by modulating oxidative stress and inflammatory pathways, positioning them as potential cross-phenotype biomarkers or therapeutic targets for aging intervention.

EGFR and NTRK3 showed context-dependent effects: both proteins were positively associated with epigenetic age measures (KDM-BA and KDM-BA acceleration), yet exhibited protective associations in clinical aging phenotypes. EGFR signaling has been implicated in the induction of cellular senescence and the senescence-associated secretory phenotype, suggesting that chronic EGFR activation may accelerate epigenetic aging despite potential benefits in tissue repair [42]. NTRK3 encodes the TrkC receptor, which activates MAPK signaling to support neuronal differentiation and survival. It plays a critical role in the development of proprioceptive neurons essential for motor coordination and cognition, suggesting a neuroprotective role in healthy aging [43]. However, NTRK3 alterations have also been linked to tumorigenesis [44], indicating context- and age-dependent functions. These dual roles may help explain its aging-preventive associations in some phenotypes and aging-promoting effects in KDM-BA and KDM-BA acceleration, which may reflect distinct epigenetic aging processes. These pleiotropic roles likely underlie the observed phenotype-specific associations, and further mechanistic studies are needed to clarify their contributions to different dimensions of aging.

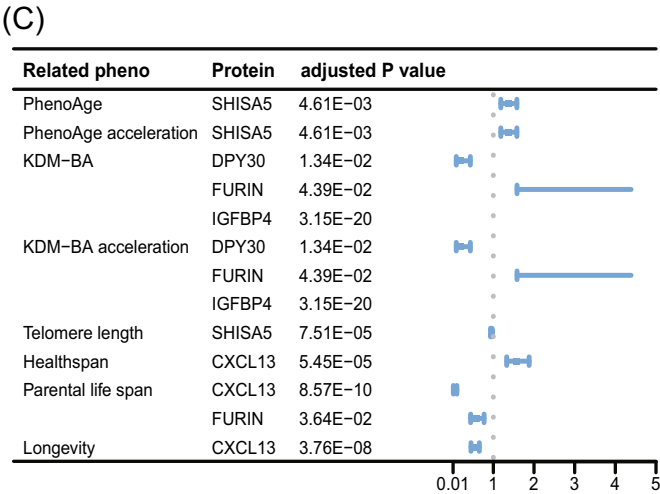
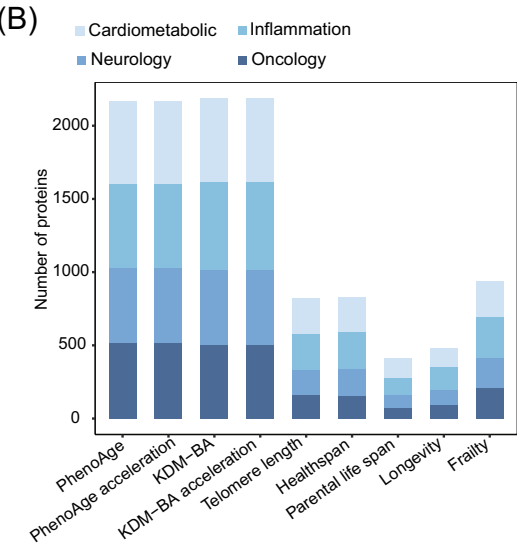
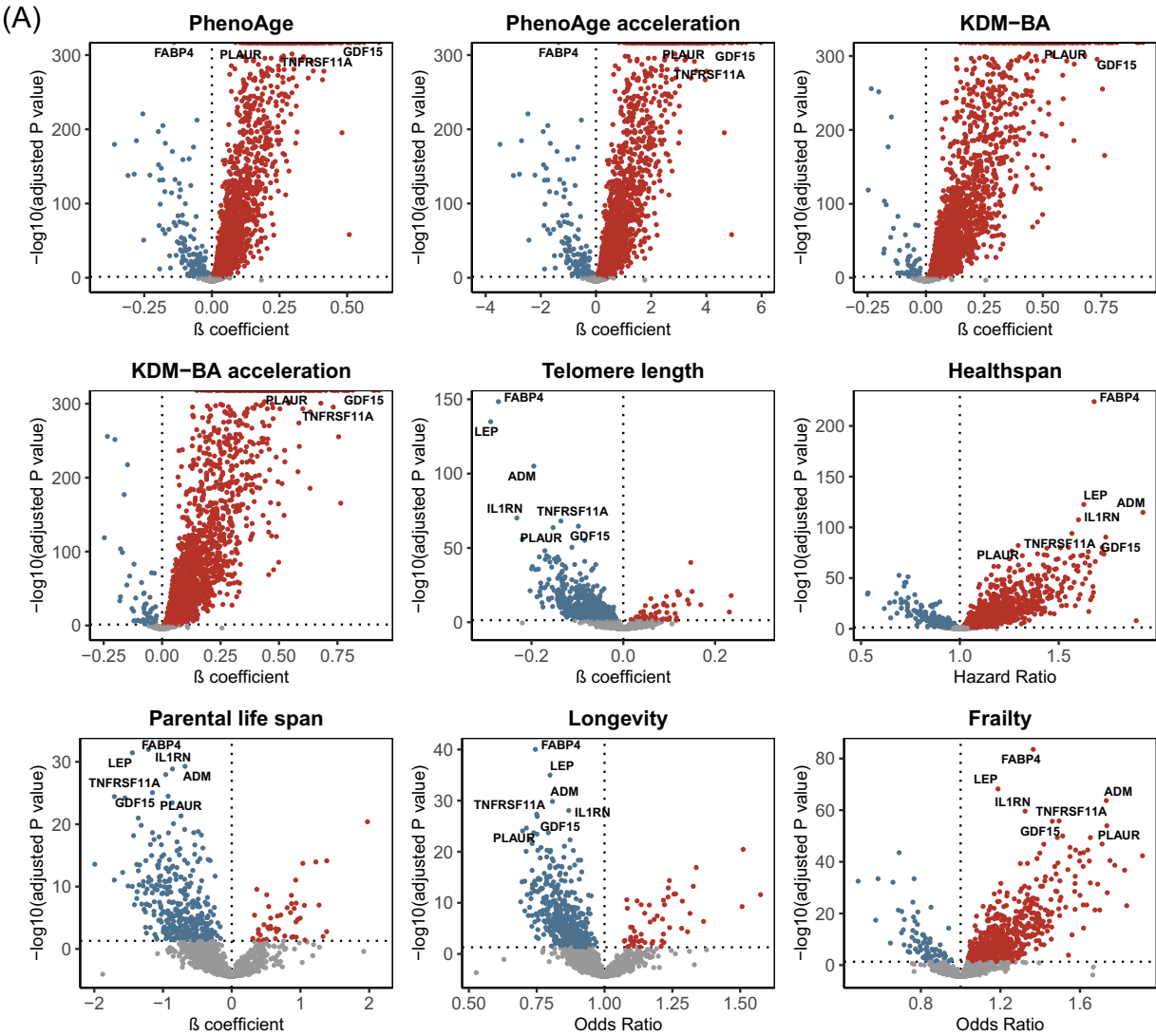
To assess the potential confounding effect of fasting duration, we performed a sensitivity analysis by additionally adjusting for fasting time in our fully adjusted model. Of the 227 originally identified proteins, 222 remained significant across all aging phenotypes (Supplementary Table 5). Five proteins (ANGPT2, EDA2R, EGFR, FOLR2, NTRK3) emerged only in the fasting-adjusted analysis, whereas five proteins (CCN3, CCN5, CXCL9, SULT2A1, THY1) were no longer significant.

To evaluate potential sex-specific differences in age-associated protein changes, we performed stratified analyses in males and females separately. A total of 94 proteins showed significant age associations in females, whereas 80 proteins were identified in males. Among these, 49 proteins were consistently associated with age in both sexes, while 45 proteins were uniquely significant in females and 31 proteins were unique to males (Supplementary Table 6–7). Notably, several immune- and inflammation-related proteins, such as CHI3L1, CD83, and HAVCR2, were consistently associated with age in both males and females, suggesting common pathways of immune aging. On the other hand, certain proteins showed sex-specific associations: for example, TGFA and LGALS9 were significantly associated with age only in females, whereas TREM2 and LEP were unique to males. These sex-specific patterns may reflect underlying differences in hormonal regulation, immune system aging, or other biological processes such as menopause. Collectively, these results underscore the importance of considering sex as a biological variable when studying age-related proteomic changes.

Biological function of aging-associated proteins

We included only those proteins that demonstrated positive results, were associated with all nine phenotypes, and exhibited consistent associations across phenotypes in the subsequent analysis ($n = 227$). Subsequently, the differential expression of the identified proteins was examined in 54 tissues using the GTEx v8 database. Our proteins exhibited significant differential expression in lung, adipose tissue, arteries, the upper gastrointestinal tract (including the esophagus and stomach), immune tissue, and the urinary system (including the kidney and bladder) compared to others (Fig. 2A). The focus of our research was on the expression of proteins in brain tissues. Notably, proteins exhibited pronounced differences in brain tissues compared to others (Supplementary Table 8). These included regions in our analysis, such as the cerebral cortex and the frontal cortex (Fig. 2A).

Functional profiling of aging-associated proteins revealed significant enrichment in inflammation-related pathways and cellular components (Supplementary Table 9, Fig. 2B). Cytokine-cytokine receptor interaction pathway exhibited strong statistical significance, alongside enrichment in the “viral protein interaction with cytokine and cytokine receptor” pathway, suggesting that age-related proteins are involved in key inflammatory signaling processes. Molecular functions such as TNF receptor activity and death receptor activity, as well as biological processes including inflammatory response, positive regulation of leukocyte migration, and leukocyte chemotaxis, were significantly overrepresented. These findings indicate that inflammatory regulation constitutes a major biological feature of aging-associated proteins, potentially contributing to the development of chronic inflammation and immune dysregulation with aging. In terms of cellular localization, aging-associated proteins were significantly enriched in components such as the endoplasmic reticulum lumen, vacuolar lumen, membrane raft, and the collagen-containing extracellular matrix. These compartments are central to protein synthesis, transport, secretion, and cell-environment interactions. The observed



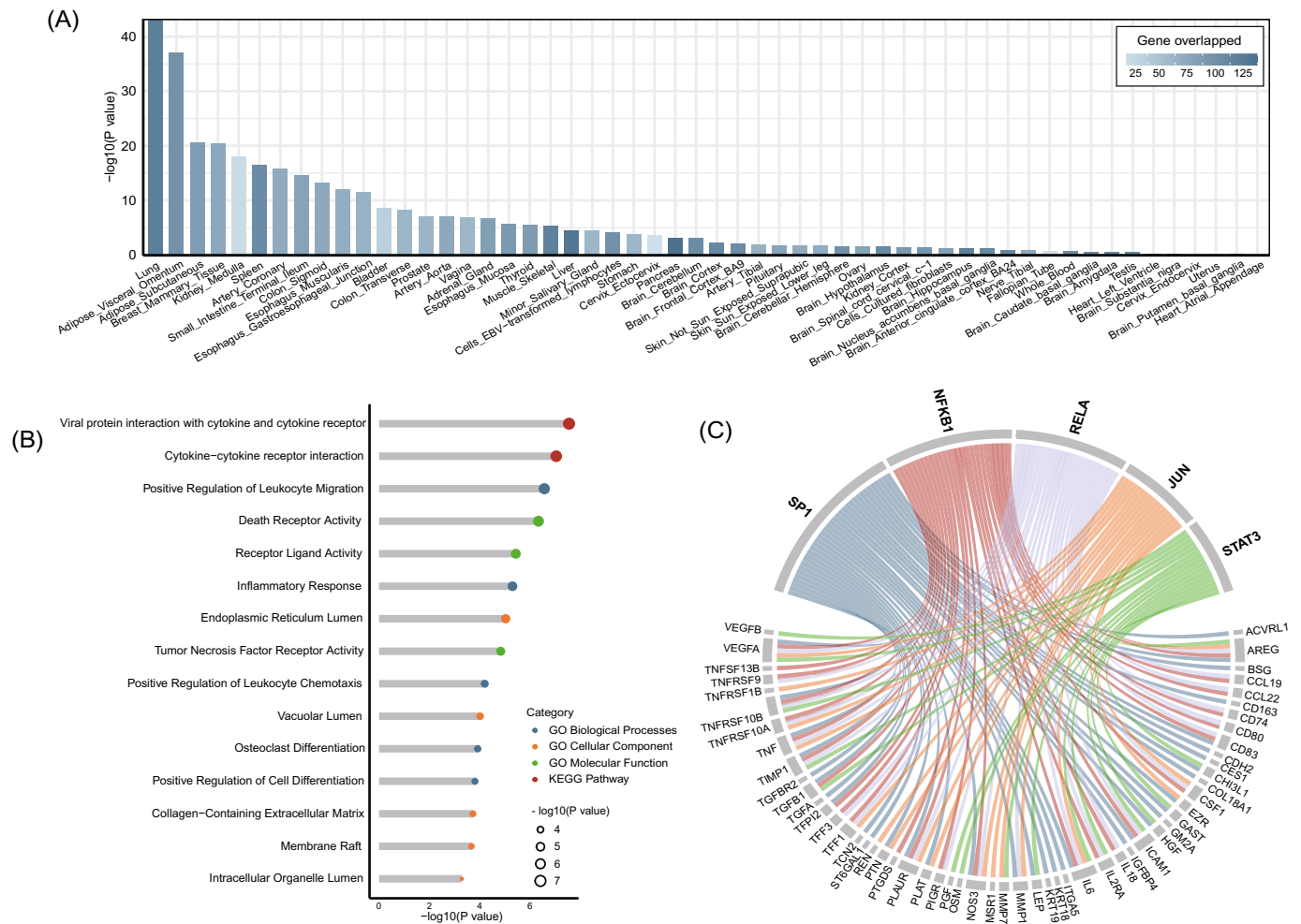


Fig. 2. Biological function of health aging-associated proteins. **A.** Plot demonstrates tissue-specific expression patterns of health aging-related proteins. Analysis was conducted using GENE2FUNC in Functional Mapping and Annotation (FUMA), based on the GTEx v8 database comprising 54 tissue types. **B.** Results of the pathway enrichment analysis. The x-axis represents the $-\log_{10}$ of the P value for each term, indicating the statistical significance after FDR correction. Different terms are listed on the y-axis, with their sources distinguished by various colors. **C.** Results of the transcription factors enrichment analysis. The transcription factors that ranked among the top 5 in significant enrichment after FDR correction are illustrated. Abbreviations: KEGG, Kyoto Encyclopedia of Genes and Genomes pathways.

changes suggest that aging alters the internal cellular environment and organelle functions, which may impair cellular metabolism, secretion, and waste clearance capacity, thereby playing a role in the initiation and progression of aging-related functional decline and diseases.

Next, we used the TRRUST database to identify potential transcription factor targets of aging-associated proteins [31]. We identified the top 5 transcription factors involved in regulating the biological effects of these genes (Fig. 2C, Supplementary Table 10). The top 5 transcription factors included transcription factor sp1 (SP1), nuclear factor (NF)- κ B subunit 1 (NFKB1), nuclear factor (NF)- κ B p65 (RELA), signal transducer and activator of transcrip-

tion 3 (STAT3), and jun proto-oncogene, AP-1 transcription factor subunit (JUN). SP1 plays a pivotal role in the process of aging, regulating nucleocytoplasmic transport genes, influencing cell signaling pathways, and being affected by oxidative stress [45]. This ultimately impacts the responsiveness of cells to external signals and their functional stability [46]. NFKB1 plays a crucial role in age-related cognitive decline through chronic inflammation [47]. In *nfk1b*^{-/-} mice with enhanced NF- κ B activity and inflammation, researchers observed early memory loss, increased neuroinflammation, and more senescent cells in specific brain areas. The administration of ibuprofen to mice with cognitive impairment resulted in a significant improvement in cognitive function,

Fig. 1. Relationship between health aging and plasma proteins. **A.** Scatter plots show the associations between 9 aging-related traits and 2,923 proteins. The Cox regression model was applied to examine the association between protein levels and healthspan. Linear regression models were applied to examine the association between protein levels and PhenoAge, PhenoAge acceleration, KDM-BA, KDM-BA acceleration, telomere length, and parental life span. Logistic regression models were applied to examine the association between protein levels and frailty and longevity. The displayed p-values were two-sided and adjusted for multiple comparisons. Proteins above the horizontal dotted black line were significantly linked to aging-related traits following Bonferroni corrections ($P < 1.90 \times 10^{-6}$). **B.** Plots show the number of each category of proteins significantly associated with aging-related traits, including four categories: cardiometabolic, inflammation, neurology, and oncology. **C.** Results of Mendelian randomization analysis on associations of plasma proteins with healthy aging. Forest plot showing causal effects of 5 MR-identified proteins on healthy aging. The boxes represent odds ratio (OR) values, and the horizontal lines represent 95 % confidence intervals.

thereby confirming the role of NFKB1 in inflammation-related cognitive decline [47]. Collectively, the TRRUST-derived transcription factors support the centrality of immune and inflammatory pathways in aging mechanisms.

Trajectories of plasma proteins during biological aging

The subsequent phase of our investigation was to ascertain the patterns of trajectory associated with plasma proteins and biological aging. We conducted a profiling of the trajectories of 227 proteins that demonstrated nominal associations with biological aging, in addition to 5 proteins of particular importance (Fig. 3A–3C). To reduce the complexity of the proteome, unsupervised hierarchical clustering was employed to group proteins with similar trajectories. This resulted in the identification of 4 clusters of protein trajectories that changed with age (Fig. 3D, Supplementary Fig. 1). The number of proteins in each cluster ranged from 2 to 107 (Supplementary Table 11). The proteins of cluster 2, 3 and 4 showed logarithmic trajectory. In general, the majority of changes observed in the plasma proteome throughout the lifespan exhibit a non-linear pattern. Out of the 4 clusters analyzed, 3 were enriched for specific biological pathways ($P < 0.05$), suggesting distinct, yet orchestrated changes in biological processes during the lifespan (Supplementary Table 12). Protein levels associated with the cytokine-mediated signaling pathway exhibited a steady increase until the biological age of 60, after which they exhibited exponential growth (cluster 2). Proteins related to the positive regulation of leukocyte migration show the fastest increase during the aging process (cluster 4).

Waves of biological aging-associated proteins

The objective of this study was to investigate the dynamic patterns of plasma proteome changes across the course of biological aging. To quantify these changes, we employed DE-SWAN analysis, focusing on two-year intervals and comparing groups within one-year parcels. The sliding window approach permitted the tracking of proteomic shifts from young to old biological ages. Notably, three significant peaks of protein changes were identified at biological ages 41, 60, and 67 (Fig. 4A–B). Even when adjusting the q-value threshold, these peaks remained detectable, underscoring the robustness of these biological age-related waves (Supplementary Table 13). For further details on the proteins associated with these peaks, please refer to Supplementary Table 14. The 5 biologically important proteins that were identified as being associated with aging yielded significant results at all three biological age waves (Fig. 4C). The analysis was repeated using KDM-BA as the biological age measure, yielding consistent results. More details can be found in the [supplementary materials](#) (Supplementary Table 15).

We identified several potentially characteristic pathways (Fig. 4D, Supplementary Table 16). We found characteristic pathways at different biological ages. At age 41, these include immune-related, metabolic, and cell signaling pathways. At age 60, they involve immune regulation, metabolic, and aging-related pathways. At age 67, pathways related to inflammation, metabolic regulation, and cell cycle regulation are characteristic. This suggests that biological age shows distinct pathway characteristics at various stages, highlighting the complexity and diversity of biological processes.

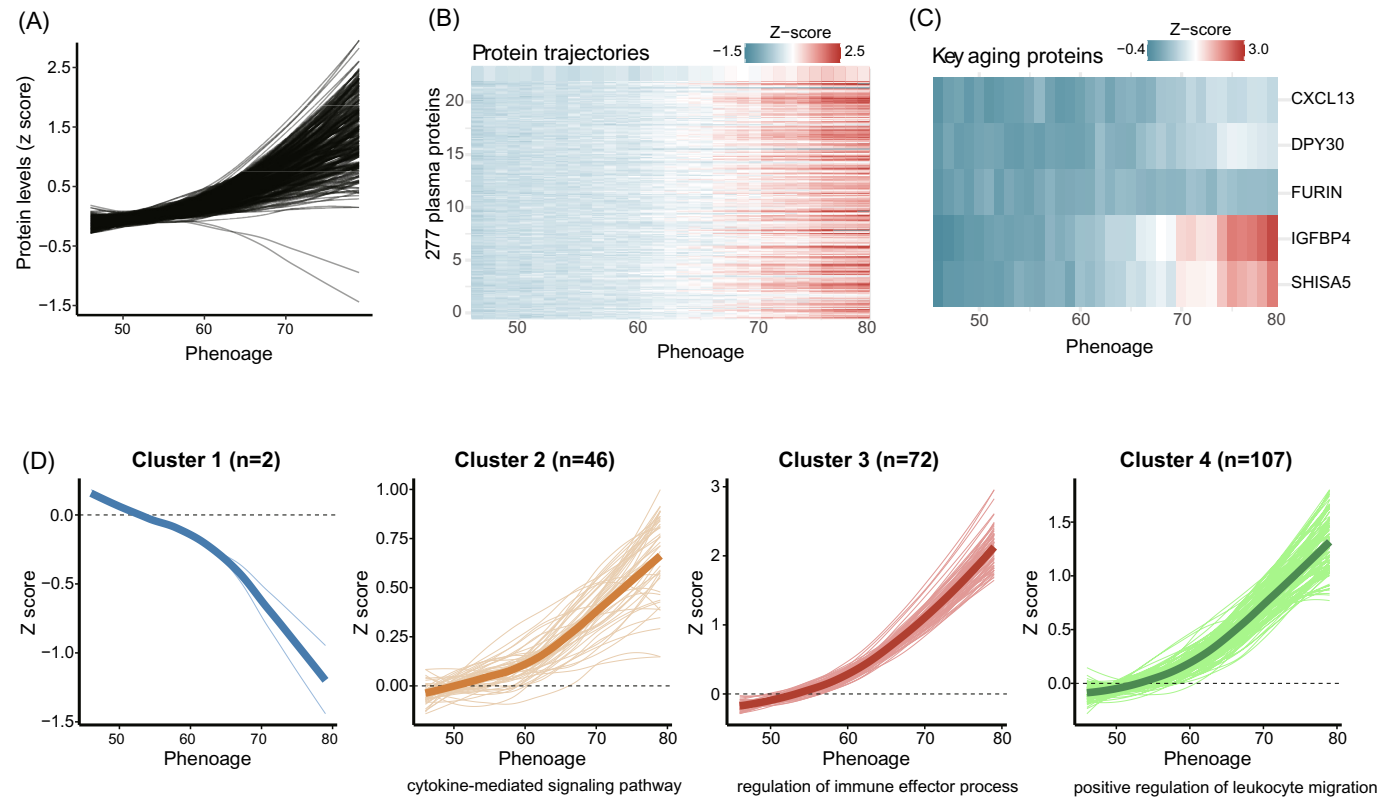


Fig. 3. Plasma proteomes trajectories during biological aging. A. The study examined plasma protein trajectories throughout the aging process. Utilizing z-score normalization, trajectories were estimated for 227 plasma proteins using the LOESS method. B. A heatmap visually displayed the trajectories of plasma proteins during the aging. C. The heatmap specifically highlighted the trajectories of 5 aging-related proteins during the aging. D. Clustering analysis identified four distinct groups of plasma protein trajectories during aging. The average trajectory of each cluster was visually emphasized with a thicker line, and the number of proteins within each cluster was annotated for reference.

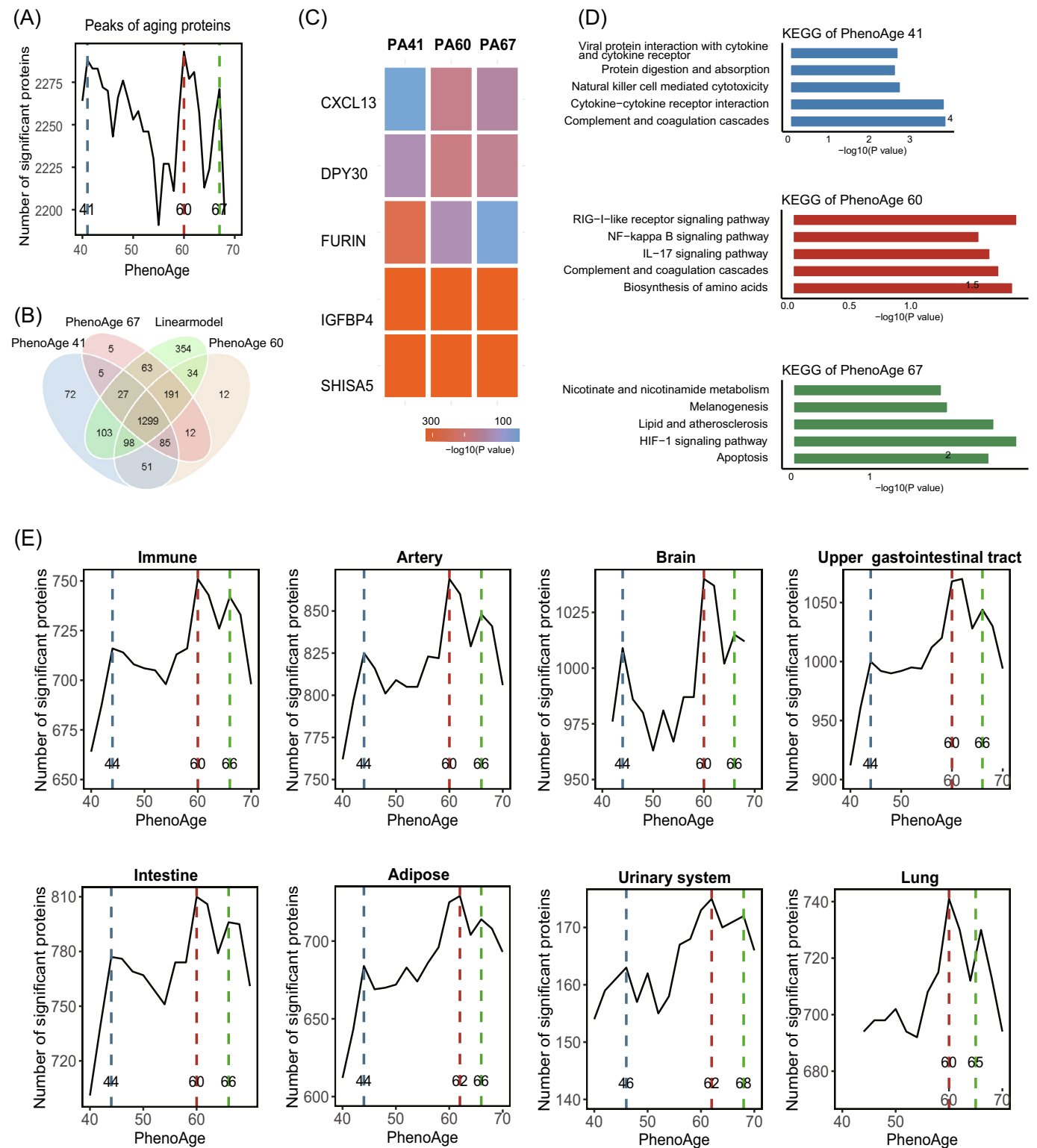


Fig. 4. Plasma protein waves during biological aging. A. The upper portion illustrates the variations in the number of plasma proteins with differential expression during the aging process. Notably, three peaks were observed at biological age 41, 60, and 67, indicating distinct phases or stages of aging. B. Overlaps between waves of biological age proteins. C. The heatmap illustrated the importance of 5 aging-related proteins at biological ages 41, 60, and 67. D. Identify the key biological processes associated with each protein wave as generated by KEGG. E. The variations in plasma protein expression during the aging process differ across various organ tissues. Except for the lungs, three peaks were observed in other organs at biological ages, indicating distinct phases or stages of aging. Notably, the first peak in the lungs appeared latest, at the age of 60.

The study discovered distinct patterns of protein changes in various organs during biological aging (Fig. 4E, Supplementary Table 13, Supplementary Table 15). Significant protein change peaks were observed at different ages for each organ, such as 44, 60, and 67 for immune, arterial, brain, upper digestive tract, and intestinal proteins. Fat tissue showed a later second peak at 62, and the urinary system's first peak appeared at 46. The lungs aged relatively later with peaks at 60 and 65. These findings not only

reveal a unique pattern of protein changes that reflect early protein changes associated with organ aging but also suggest a shared mechanism of aging across different organs. This shared susceptibility to protein-related biological processes allows for simultaneous monitoring of organ aging processes and related physiological changes.

Sex-specific waves of aging-associated plasma proteins

To explore sex-specific trajectories of proteomic aging, we performed DE-SWAN analysis stratified by sex (Supplementary Table 17). In females, we observed three distinct peaks in aging-associated protein changes, occurring at biological ages 49 and 64 (Supplementary Fig. 3). Notably, the trajectory after age 49 revealed a sharp, sustained decline in the number of significant proteins, suggesting a potential inflection point in systemic aging. This decline may reflect the abrupt physiological changes associated with perimenopause and menopause, including hormonal withdrawal, altered immune regulation, and increased inflammatory burden. Although further peaks at 64 was detected, it was less pronounced and occurred on a declining background trend, highlighting a potential “cliff-like” trajectory of aging in females.

In contrast, males exhibited a different pattern, with proteomic aging following a more oscillatory trajectory. Three peaks were observed at biological ages 46, 56, and 67 (Supplementary Fig. 4), without a comparable post-peak decline as seen in females. These fluctuations may reflect a more gradual accumulation of age-related molecular changes, potentially modulated by sex-specific differences in hormonal decline, metabolic processes, and immune aging. The third peak at 67 overlapped with that of the total population, suggesting a common late-life aging signature across sexes. These findings emphasize the divergent trajectories of biological aging between men and women and underscore the necessity of sex-stratified analyses in aging research.

Drug target implications

Two-sample Mendelian randomization (MR) analyses were conducted to explore a potential causal link between 227 aging proteins and 9 associated phenotypes. A total of 6,278 blood *cis*-pQTLs for 225 proteins were available for analysis (Supplementary Table 18). Our primary analysis revealed 5 proteins in plasma associated with aging risks (Fig. 1C, Supplementary Table 19). Genetically higher levels of plasma DPY30, FURIN, and IGFBP4 were linked to increased risks of KDM-BA and its acceleration, while elevated levels of SHISA5 was associated with higher risks of PhenoAge and its acceleration. The results indicated that genetically predicted plasma CXCL13 levels were associated with an increased risk of the end of healthspan ($\beta = 0.458$, $P = 2.42E-07$), shorter longevity ($\beta = -0.610$, $P = 1.67E-10$), and a shorter parental lifespan ($\beta = -3.171$, $P = 3.81E-12$). No associations were identified between the other aging proteins and aging (all $P > 0.05$). The F statistic for all instruments ranged from 13 to 20,740 (Supplementary Table 18). We performed Steiger filtering to confirm that all causal directions were from protein levels to aging-related traits (Supplementary Table 20), supporting the robustness of our MR findings. Furthermore, it was found that ten proteins are druggable. The druggability tier of each protein is detailed in Supplementary Table 21.

Results of phenome-wide association analysis

We examined the relationship between aging proteins and organ-related phenotypes (Supplementary Table 22). After Bonferroni correction, 103 significant associations were identified (Fig. 5A and Supplementary Table 23). DPY30 was associated with the

highest number of phenotypes, specifically those linked to inflammation. The most frequent associations were noted in relation to inflammation (such as CRP, eosinophils, lymphocytes, monocytes, white blood cells, neutrophils), adipose tissue (trunk fat mass and whole-body fat mass), and blood components (like LDL cholesterol and triglycerides), which were connected to 5 proteins. Not each protein plays a significant role in these cognitive aspects. IGFBP4, CXCL13, and SHISA5 are closely associated with cognition, particularly in numeric memory and reaction time. The correlation between aging-related proteins and phenotypes across organs underscores their vital roles in biological processes. These proteins likely regulate overall health, reflecting inter-organ interactions and communication networks. This broad impact indicates their crucial role in maintaining the balance and health of multiple organs.

We have further characterized the disease relevance and clinical significance of the identified proteins. After Bonferroni correction, 82 significant associations were identified (Fig. 5B and Supplementary Table 23). All 5 proteins (CXCL13, DPY30, FURIN, IGFBP4, SHISA5) are strongly linked to various diseases and mortality. They correlate significantly with cardiovascular diseases, neurological disorders, cancer, digestive and urinary system diseases, respiratory disorders, and mortality. Cardiovascular diseases, particularly myocardial infarction, congestive heart failure, hypertension, and angina pectoris, have the highest frequency of associations among these proteins, with links to 5 proteins. Research on the impact of 5 proteins on mortality revealed significant findings. SHISA5 posed the highest risk (HR = 2.454), followed by IGFBP4 (HR = 2.031), FURIN (HR = 1.346), DPY30 (HR = 1.314), and CXCL13 (HR = 1.312). These data indicate age-related changes in specific proteins affecting multiple organs and disease risks, especially cardiovascular metabolism. This underscores the importance of further aging research, suggesting promising investigative directions.

The impact of risk factors on the aging process is mediated by plasma proteins

We are interested in exploring whether cardiovascular metabolic factors can influence the aging process by modulating key proteins associated with aging, especially given the observed correlation between the comprehensive cardiovascular metabolism score LE8 and aging [48] (Fig. 5C). Among the 45 potential triangles that encompass significant and consistent directional connections within clinical factor-protein associations related to healthy aging, a total of 17 triangles displayed notable mediation effects (Supplementary Table 25). We identified five proteins with MR-positive association with aging, acting as mediators in the relationship between LE8 and healthy aging (Fig. 5D). The median percentage of mediation among all significant triangles was 41.99 %. We observed that SHISA5 is a significant mediator in multiple biological processes, exhibiting the most pronounced positive mediating effects across various outcomes. Specifically, SHISA5 demonstrates significant positive mediation in frailty (48.5 %), KDM-BA (10.0 %), KDM-BA acceleration (10.0 %), PhenoAge (46.1 %), PhenoAge acceleration (45.7 %), and healthspan (21.2 %).

Discussion

Plasma proteomics is a promising method for identifying clinical biomarkers associated with aging. In this study, we analyzed 2,923 proteins from the UKB to investigate their associations with 9 aging-related traits: PhenoAge, PhenoAge acceleration, KDM-BA, KDM-BA acceleration, healthspan, LTL, parental lifespan, longevity, and frailty. A total of 227 proteins were consistently associated

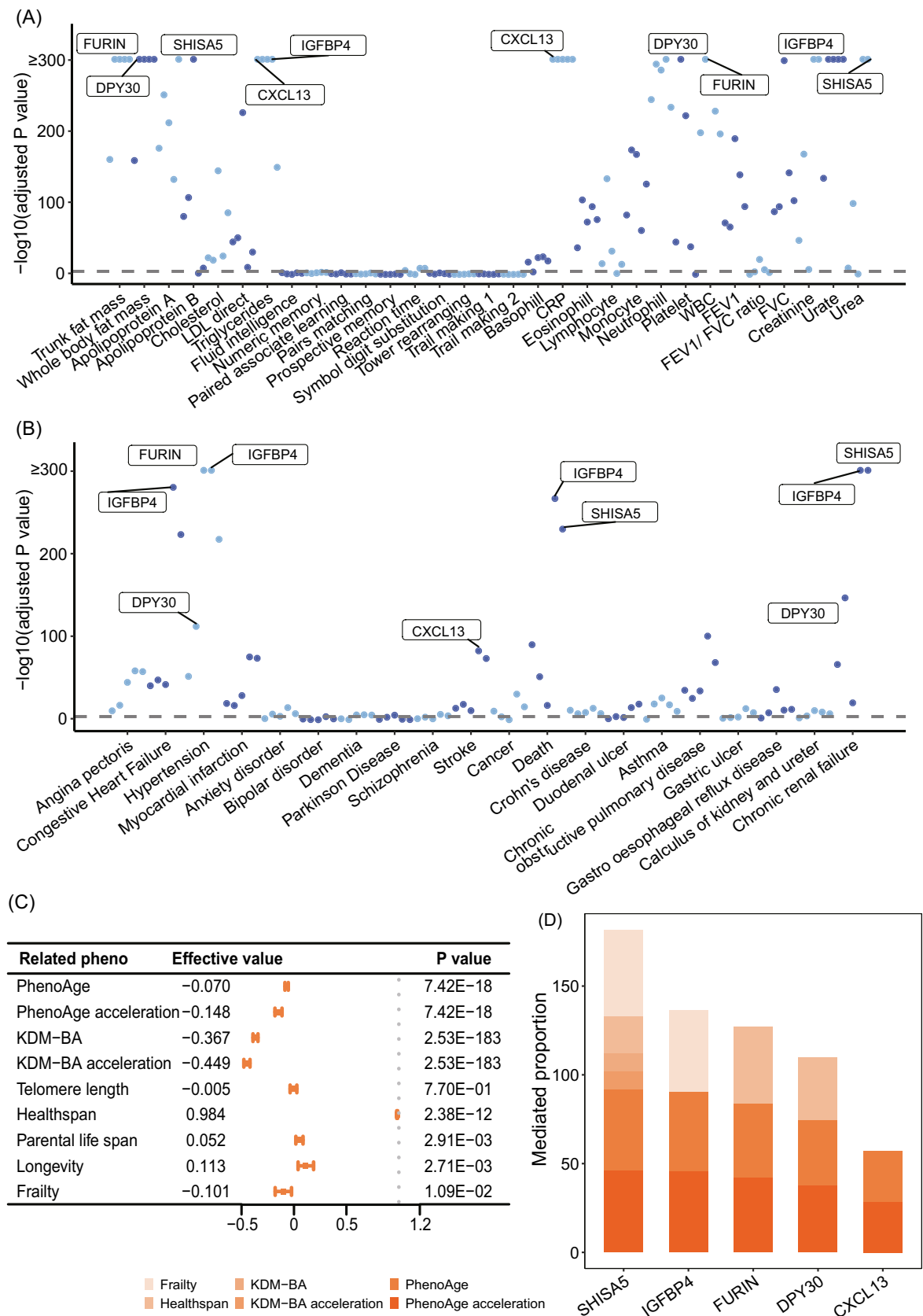


Fig. 5. Phenome-wide association analysis of aging-associated proteins. A-B. phenotype-wide association analysis between 5 key aging-related proteins and organ phenotypes, diseases, and mortality. The y-axis indicates the $-\log_{10}$ of the P values for each association, and the x-axis represents different phenotype categories. The P values shown are two-sided and adjusted for multiple testing. Grey line in each figure is Bonferroni 0.05 correction threshold. C. The association between LE8 and aging. D. The bar chart displays the proportion of mediating proteins and incident outcomes for each exposure. This indicates the proportion of impact of modifiable risk factors on aging mediated through protein intermediates. Detailed median proportion significantly mediated by each protein across health aging is provided.

with all phenotypes. Further investigation was conducted into the biological functions and implications for aging of these proteins. Additionally, our study revealed dynamic changes during biological aging, with proteomic alterations peaking in the late 41, 60, and late 67 decades of biological aging. These findings highlight these periods as crucial for intervening in the biological aging process. Furthermore, we tested their causal associations with aging, identifying CXCL13, DPY30, FURIN, IGFBP4, and SHISA5 as promising candidate biomarkers for aging. The results of the PheWAS analysis for proteins associated with aging indicate that they play an essential role in biological processes, including the regulation of overall health and the maintenance of balance across multiple organs.

Our research emphasizes the crucial role of inflammation in aging. The key proteins associated with aging were found to be primarily enriched in inflammation-related pathways. Moreover, following the clustering of these proteins, it was observed that several clusters' key pathologies were also associated with inflammation. Recent research has identified the potential of targeting the pro-inflammatory cytokine network as a strategy to combat aging. Anti-inflammatory drugs, including metformin, aspirin, rapamycin, and ibuprofen, are currently under investigation for their potential anti-aging effects. For instance, metformin has demonstrated efficacy in reducing chronic inflammation and promoting healthy aging by modulating pathways such as IKK/NF- κ B in individuals with Type 2 diabetes [49], as well as GPX7/NRF2 [50] and the recently identified target, PEN2 [51]. Aspirin has been demonstrated to delay replicative senescence by reducing oxidative stress. Moreover, studies have identified CD36 as a pivotal mediator of SASP-related mechanisms. CD36-specific short interfering RNA has been demonstrated to effectively reduce SASP secretion in senescent muscle tissue cells [52].

Our study demonstrated that aging-associated proteins exhibit tissue-specific expression patterns across multiple organs, with particularly pronounced enrichment observed in certain regions, such as the cerebral and frontal cortex of the brain. This supports growing evidence that distinct molecular processes underlie organ-specific aging trajectories. In the brain, regions especially vulnerable to aging—such as the prefrontal cortex—undergo marked alterations in synaptic plasticity and ion channel homeostasis. For instance, SYNPR, a synaptic vesicle-associated protein involved in modulating chemical synaptic transmission, has been associated with A β -induced impairment of acetylcholine (ACh) release in AD [53]. Similarly, GRIN2A, a subunit of the N-methyl-D-aspartate (NMDA) receptor that governs neuronal excitability and long-term potentiation, exhibited aberrant splicing patterns in the aging cortex [54]. Dysregulation of GRIN2A has been linked to calcium-mediated excitotoxicity in aged neurons, thereby exacerbating oxidative stress and neurodegeneration in Parkinson's disease models [55]. These findings underscore the relevance of tissue-specific protein alterations—particularly in the brain—in mediating structural and functional decline during aging. Notably, while the brain remains a central focus given its vulnerability to age-related cognitive deterioration, our identification of protein enrichment across multiple organs suggests that distinct molecular signatures may underpin organ-specific aging trajectories. Investigating these tissue-enriched proteins may not only advance our understanding of multi organs, but also offer novel biomarkers and mechanistic insights into the aging processes of different organs.

Another key finding of our study is the undulating plasma proteomic changes that occur throughout the process of biological aging. Three essential protein waves associated with biological age at 41, 60, and 67 years were identified, coinciding with dramatic physiological changes. The proteins present at these peaks exhibit partial overlap in terms of their composition and functional

enrichment. In contrast to the general biological aging process, our findings indicate that 41 may represent a potential onset time point for biological aging, characterized by immune-related, metabolic, and cell signaling pathways, which was a potential mechanism of aging during this period. The 60-year-old state coincides with the menopausal transition. The changes that occur during this period are more pronounced than those observed in other phases of life [56]. These changes are not only evident in hormone levels but also in immunity and tissue function. In addition, we found that proteins associated with the HIF-1 signaling pathway were differentially expressed at biological age 67 years. Research indicates that the HIF-1 signaling pathway, as well as factors such as p53 and TNF- α , are critical in cardiovascular conditions like arterial dissection, atheroma, and atherosclerosis [57,58]. HIF1A is involved in both the HIF-1 signaling pathway and the p53 signaling pathway, resulting in the upregulation of p53 via HIF-1 α under hypoxic conditions [59]. Overall, these findings indicate that the pace of biological aging is uneven across the lifespan, and further studies are warranted to elucidate the underlying mechanisms of certain proteins at different stages of biological aging.

In our analysis, we identified several potential biomarkers for biological aging. CXCL13, DPY30, FURIN, IGFBP4, and SHISA5 emerged as a key potential drug target with the highest importance score, as indicated by the MR. The role of CXCL13 in aging is primarily evident in neuroinflammation and immune responses. Studies have shown that with aging, the regeneration of microglia leads to increased expression of CXCL13, which is associated with changes in tau protein in AD [60]. Additionally, in immune checkpoint inhibitor therapy, CXCL13 plays a role by attracting B cells to damaged organs, contributing to immune-related adverse events [61]. These findings suggest that CXCL13 is crucial in regulating neuroinflammation and immune responses related to aging. DPY30, a core component of H3K4 histone methyltransferase complexes, has been identified as a crucial regulator of cellular proliferation and senescence [62]. Loss of DPY30 induces a senescent phenotype characterized by increased ROS, activation of CDK inhibitors such as p16INK4a, and transcriptional dysregulation of cell cycle genes via ID protein pathways, suggesting its central role in modulating aging-related cellular stress responses [62]. The role of FURIN in aging is primarily reflected in its changes in expression and localization. Under diabetic conditions, FURIN expression in glomerular cells decreases, but it accumulates in the endoplasmic reticulum and Golgi apparatus, leading to dysregulated extracellular matrix remodeling. This dysregulation accelerates the deposition of GBM material, impairs glomerular filtration function, and consequently hastens the aging process [63]. IGFBP4 is released by senescent cells as part of the senescence-associated secretory phenotype and acts as a stress mediator. Genotoxic stressors like low-dose irradiation can trigger its release into the bloodstream, linked to aging effects. Injecting IGFBP4 into mice increased senescent cells in their lungs, heart, and kidneys [64]. SHISA5, also known as SCOTIN, is a p53-inducible transmembrane protein localized to the endoplasmic reticulum and nuclear membrane, and is known to mediate apoptosis under cellular stress condition. p53 plays a crucial role in the aging process, and its functional decline is associated with increased cancer incidence in older populations [65]. Moreover, different p53 isoforms have been shown to regulate cellular senescence and age-related diseases [66]. These findings suggest that SHISA5 may be involved in aging through the p53 signaling pathway by modulating cell survival and stress responses. Understanding the impact of these 5 proteins on aging and metabolic diseases like diabetes and AD offers opportunities for targeted interventions and therapies.

These proteins significantly affect the function of multiple organs, emphasizing the importance of objective assessment and avoiding biased language. The study participants were young indi-

viduals from the UKB, representing early stages of aging. Among these younger cohorts, the research revealed a distinct pattern of protein changes that more accurately reflected early aging-related protein alterations. The study identified various proteins associated with lung function, inflammation, and brain function. These findings support the concept of aging occurring concurrently in different organs and potentially sharing common susceptibility to protein-related biological processes. This shared mechanism enables the simultaneous monitoring of aging processes and related physiological changes across various organs.

This study combines high-throughput proteomic profiling with in-depth assessments of healthy aging in a large community-based cohort, complemented by extensive long-term clinical follow-up. Previous proteomic studies in the UK Biobank have largely focused on individual aging phenotypes. For example, Xue et al. assayed 2,920 plasma proteins in 43,895 participants and identified 102 and 90 proteins associated with prefrailty and frailty [67]. Similarly, Zhao et al. applied proteome-wide Mendelian randomization to 2,923 proteins and telomere length GWAS data in UKB, identifying 22 proteins causally linked to leukocyte telomere length [68]. While these studies provided valuable insights into specific aging-related traits, their single-phenotype approach may overlook shared biological mechanisms across aging dimensions. In contrast, our study leverages validated biological age measures (KDM-BA and PhenoAge) and systematically examines 2,923 proteins across nine distinct aging phenotypes in over 50,000 individuals. This multi-dimensional approach revealed 12,193 Bonferroni-significant associations, far exceeding the scope of prior single-trait analyses. Notably, 227 proteins showed consistent associations across all outcomes, highlighting their potential as systemic aging biomarkers. By integrating molecular, clinical, and functional aging dimensions, we not only replicate previously reported associations (e.g., inflammatory and oxidative-stress pathways) but also uncover novel cross-phenotype regulators such as PON3 and UMOD. These findings position our study as a comprehensive resource for understanding shared aging mechanisms and identifying high-priority therapeutic targets for healthy aging interventions.

However, it is important to note some limitations when interpreting the results. Firstly, the current platform used by Olink may not fully cover the human proteome due to potential biases in protein measurement preferences, despite providing a comprehensive measurement of circulating plasma proteins. However, our study successfully addressed the scientific questions we set out to investigate by using data from over 50,000 individuals and measuring 2,923 proteins. It is important to note that most of our study participants are Caucasian. Although we have performed internal and external cross-validation and observed well-calibrated models, it is critical to validate our research findings in larger, ethnically diverse populations. Third, it is possible that individuals who participate in the UKB are more health-conscious than those who do not, which could lead to underestimations of healthspan prevalence and incidence rates. However, previous research has shown a strong correlation between identified cases of chronic disease and primary care records.

Conclusion

Our study illuminates the proteomic alterations in plasma that occur during biological aging. Notably, we identified biological age 41, 60, and 67 as pivotal transition points and identified GDF15, CXCL13, DPY30, FURIN, IGFBP4, and SHISA5 as promising biomarkers for biological aging. These findings contribute significantly to our understanding of the molecular mechanisms underlying biological aging, with implications for the development of systemic

biomarkers and personalized therapeutic targets for age-related disorders in the future.

Data availability

The data used in this study (including phenotypic and genetic data at the individual level) were primarily obtained from the UK Biobank, application number 19542, available via the UK Biobank (<https://www.ukbiobank.ac.uk/>).

Ethics approval and consent to participate

The study was conducted following the Declaration of Helsinki. The UK Biobank has research tissue bank approval from the North West Multi-Center Research Ethics Committee (11/NW/0382). Written informed consent was obtained from all participants. The present study was approved by UK Biobank under application number 19542.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jare.2025.05.004>.

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